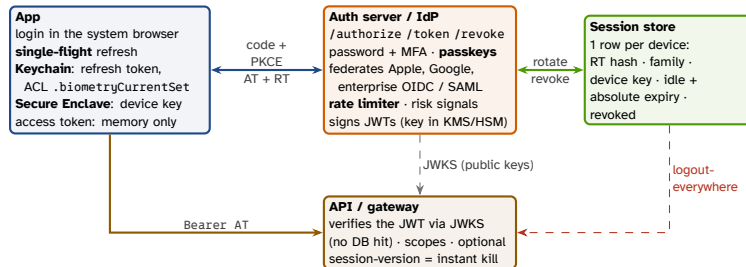
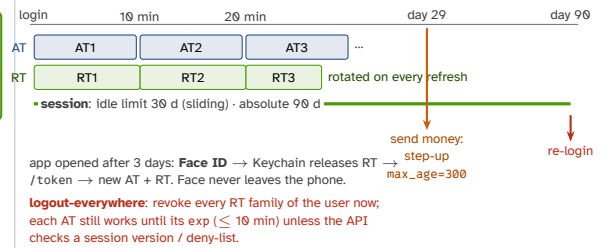


In one line: Login is a **system**, not a screen. An identity step proves *who* (password + MFA, **passkey**, SSO); the server issues a **short-lived access token** and a **server-side, revocable refresh credential** bound to *this device*, which the app keeps in the Keychain and re-opens with **biometrics**. Refresh, step-up, recovery, logout and rate limits need the same care as the happy path — attackers pick the weakest one.

The boxes



Token lifecycle (example policy)



Session vs token — the real choice

	server session	self-contained JWT
credential	opaque random id	signed claims
per request	store lookup	signature check only
revoke	delete the row; instant	wait for exp
scales by	shared store	stateless verifiers

Mobile default = **hybrid**: short JWT *access* (stateless, 5–15 min) + opaque *refresh* stored server-side = the **revocation point**.
Web: HttpOnly cookie session or a BFF — no tokens in JS.

Re-authentication with biometrics — two designs

- A. Biometry-gated secret:** RT in the Keychain with SecAccessControl .biometryCurrentSet; Face ID makes securityd release it, then a normal refresh. Simple; re-enrolment destroys the item → full login.
- B. Device key + challenge** (stronger): at login the app registers a Secure Enclave P-256 public key; unlock = server nonce → SE signs after Face ID → server verifies → tokens. Proves possession *and* presence; a copied RT alone is useless.
- Either way biometry is *local*; the server trusts a key, never a Bool. Fallback: passcode (.userPresence) or full login. touchIDAuthenticationAllowableReuseDuration lets a recent biometric *device unlock* count — not an earlier in-app check.

Device binding & step-up

Bind tokens to a key the device holds — **DPoP** (RFC 9449) proofs, mTLS, or design B — so a leaked token can't be replayed elsewhere; add App Attest / Play Integrity to vouch for the *app*.
Step-up for money, email or password change: OIDC max_age / prompt=login / acr_values, check auth_time; APIs answer insufficient_user_authentication (RFC 9470).

Example — the session store is the revocation point

```
CREATE TABLE session (
  id uuid PRIMARY KEY, user_id uuid NOT NULL,
  rt_hash bytea NOT NULL, -- never the raw token
  family uuid NOT NULL, -- reuse -> kill family
  device_pubkey bytea, -- design B / DPoP
  last_used timestampz, absolute_exp timestampz,
  revoked_at timestampz);
-- log out everywhere (also on password reset):
UPDATE session SET revoked_at = now()
WHERE user_id = $1 AND revoked_at IS NULL;
```

Remember

“Short access, revocable refresh, bound to the device, opened by biometrics — and recovery as strong as login.”

Identity step — passkeys, SSO, MFA

- Passkeys** first: WebAuthn key pair per site, phishing-resistant, nothing to steal server-side. iOS: AutoFill sign-in (16) · auto-upgrade after a password login (18) · passkey sign-up ASAuthorizationAccountCreationProvider (26). Keep recovery.
- SSO**: social or enterprise — your IdP **federates** (OIDC/SAML) and still issues *your* tokens; the app talks only to your auth server (ASWebAuthenticationSession).
- MFA strength**: passkey / security key (phishing-resistant) > TOTP (RFC 6238, 30 s) > push (**number matching** vs fatigue) > SMS OTP (SIM swap; NIST “restricted”).

Brute force & rate limiting

Limit per **account** *and* per IP/device, plus globally; progressive delays + risk challenges, not hard lockout. **Credential stuffing** (breached pairs, one try per account) beats per-account limits — OWASP: MFA is “by far the best defense”. Same answer and timing for “no such user” and “wrong password”. 429 (+ Retry-After). **NIST SP 800-63B-4** (Aug 2025): ≥ 15 chars if the password is the only factor (≥ 8 with MFA), *no* composition rules, *no* periodic change, check a breached/common blacklist, ≤ 100 consecutive failures per account.

Recovery & logout

Recovery is the real front door: single-use short-lived email link or recovery codes, a second factor if enrolled, a cool-down, a notice to every device; never weaker than login, never reveals whether an account exists. **Logout**: /revoke the RT (RFC 7009) + wipe Keychain + end the IdP session (else the next login is silent). **Everywhere**: revoke all rows, bump the session version; OIDC back-channel logout tells relying parties.

Traps

- ! “JWTs can’t be logged out” — put the kill at the refresh token; keep ATs short.
- ! Face ID as if ok { showHome() } — gate a key or a secret, not a screen.
- ! Password reset that leaves old sessions alive — the attacker stays in.
- ! Hard lockout after 5 tries = a denial-of-service button.